# Solve with us 8 - Not Graded

This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- Two non-zero vectors $u$u and $v$v from an inner product space $V$V are said to be orthogonal if $\langle u, v\rangle = 0.\langle \mathrm{u}, \mathrm{v}\rangle = 0$.
- A set $S$S of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set $S$S are mutually orthogonal. i.e., $\langle v_i, v_j\rangle = 0\langle \mathrm{vi}, \mathrm{vj}\rangle = 0$ for $i \neq j$i $\neq$ j, any $v_i, v_j \in S$vi, vj $\in$ S.
- Let $\{v_1, v_2, \ldots, v_n\}\{\mathrm{v1}, \mathrm{v2}, \ldots, \mathrm{vn}\}$ be an orthogonal set of vectors in an inner product space. Then $\{v_1, v_2, \ldots, v_n\}\{\mathrm{v1}, \mathrm{v2}, \ldots, \mathrm{vn}\}$ is a linearly independent set of vectors.
- Orthogonal basis: Let $V$V be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.
- An orthonormal set of vectors of an inner product space $V$V is an orthogonal set of vectors such that the norm of each vector of the set is 1.
- An orthonormal basis is an orthonormal set of vectors which form a basis.
- Obtaining orthonormal set from orthogonal set:Obtaining orthonormal set from orthogonal set: If $\gamma = \{v_1, v_2, \ldots, v_n\}\gamma = \{\mathrm{v1}, \mathrm{v2}, \ldots, \mathrm{vn}\}$ is an orthogonal set of vectors then we can obtain an orthonormal set of vectors $\beta\beta$ from $\gamma\gamma$ by dividing each vector by its norm i.e.,
$$\beta = \{\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \ldots, \frac{v_n}{\|v_n\|}\}.\beta = \{\|\mathrm{v1}\|\mathrm{v1}, \|\mathrm{v2}\|\mathrm{v2}, \ldots, \|\mathrm{vn}\|\mathrm{vn}\}.$$

Consider a set of vectors
$S = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}\mathrm{S} = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, 21, -23), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$ from the inner product space $R^3$ R3 with usual inner product (i.e., the dot product).
Answer questions (1) and (2).

*1 point*
Which of the following subsets of the set $S$S consist of mutually orthogonal vectors?

- [ ] $\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}$
- [ ] $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}\{(1, -1, -1), (-1, 21, -23)\}$
- [ ] $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2}), (1, 2, 0)\}\{(1, -1, -1), (-1, 21, -23), (1, 2, 0)\}$
- [ ] $S$S
- [ ] $\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}\{(1, -1, -1), (-1, 21, -23)\}$
$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

*1 point*
Which of the following option(s) is (are) true?

- [ ] $\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to $R^2$.
- [ ] $\{(2,1,1)\}$ is not an orthogonal basis of a vector space which is isomorphic to $R$.
- [ ] $\{(1,-1,-1),(1,2,0)\}$ is an orthogonal basis of a vector space which is isomorphic to $R^2$.
- [ ] $\{(-1,1,1),(-1,\frac{1}{2},-\frac{3}{2}),(0,3,1),\}$ is an orthogonal basis of $R^3$.
- [ ] $\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of $R^3$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to $R^2$.
$\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of $R^3$.

Key Points: 2

- The projection of a vector to a subspace: Let $V$ be an inner product space, $v \in V$ and $W \subseteq V$ be a subspace. Then the projection of $v$ onto $W$ is the vector in $W$, denoted by $proj_W(v)$, computed as follows:

  Find an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ for $W$.

  Then $proj_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$.

  Remark: $proj_W(v)$ is independent of the chosen orthonormal basis.
- Let $V$ be an inner product space and $W$ be a subspace. Then the projection of vectors in $V$ to $W$ is a linear transformation from $V$ to $V$ with image $W$.

*1 point*
Consider the vector space $R^3$ with usual inner product. Which of the following vectors represents projection of the vector $u = (2,1,-2)$ onto the subspace whose an orthonormal basis is

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\}\{(\quad 2$
$\sqrt{}$
$1, 0, \quad 2$
$\sqrt{}$
$1), (3-2, 31, 32)\}$?

- $(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$
- (1,1,1)
- (2,1,-2)
- $(\sqrt{2}, 0, \sqrt{2})(\quad 2$
$\sqrt{}$
$, 0, \quad 2$
$\sqrt{}$
$)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$

Consider the vector space $R^3$ R3 with usual inner product. If $(a, b, c)$(a, b, c) is the projection vector obtained from projecting the vector $(1, 0, 1)$(1, 0, 1) onto the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$$\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$, then find the value $3(a + b + c)$3(a + b + c).

Hint:Hint: Find any orthonormal basis of the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$ $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$ and use the definition of the projection of a vector onto a subspace.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 4
*1 point*

Key Points: 3Key Points: 3

- Gram-schmidt process:Gram-schmidt process: The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.
- Let $\gamma = \{v_1, v_2, \ldots, v_n\}$γ = {v1, v2, … , vn} be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis $\beta = \{u_1, u_2, \ldots, u_n\}$β = {u1, u2 , … , un}.
  Step-1 $w_1 = v_1$ w1 = v1 , and $u_1 = \frac{w_1}{\|w_1\|}$u1 = ‖w1‖w1.
  Step-2 $w_2 = v_2 - \langle v_2, u_1\rangle u_1$ w2 = v2 − ⟨v2, u1⟩u1 and $u_2 = \frac{w_2}{\|w_2\|}$u2 = ‖w2‖w2.

  Step-3 $w_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$ w3 = v3 − ⟨v3, u1⟩u1 − ⟨v3, u2⟩u2 and $u_3 = \frac{w_3}{\|w_3\|}$u3 = ‖w3‖w3.

$\vdots \quad \vdots$

Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ wi = vi − ∑j=1i−1⟨vi, uj⟩uj and $u_i = \frac{w_i}{\|w_i\|}$ui = ‖wi‖wi.

$\vdots \quad \vdots$

Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ wn = vn − ∑j=1n−1⟨vn, uj⟩uj and $u_n = \frac{w_n}{\|w_n\|}$un = ‖wn‖wn.

Let $\beta$β be an ordered basis of the inner product space $R^3$R3 with usual inner product, where $\beta = \{(1, 1, -1), (1, 1, 0), (1, 0, 0)\}$β = {(1, 1, −1), (1, 1, 0), (1, 0, 0)} and $\gamma = \{u_1, u_2, u_3\}$γ = {u1, u2, u3} be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

If $u_1 = (a_1, b_1, c_1)$u1 = (a1, b1, c1), then find the value of $\sqrt{3}(a_1 + b_1 + c_1)$. 3

√

(a1 + b1 + c1).

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*
If $u_2 = (a_2, b_2, c_2)$u2 = (a2, b2, c2), then find the value of $\sqrt{6}(a_2 + b_2 + c_2)$. 6

√

(a2 + b2 + c2).

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 4
*1 point*
If $u_3 = (a_3, b_3, c_3)$u3 = (a3, b3, c3), then find the value of $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$. 2

√

1(a3 + b3 + c3).

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*

Key Points: 6Key Points: 6

- Orthogonal transformation:Orthogonal transformation: Let $V$V be an inner product space and $T$T be a linear transformation from $V$V to $V$V . $T$T is said to be an orthogonal transformation if

$$\langle Tv, Tw \rangle = \langle v, w \rangle, \ \forall v, w \in V \langle \mathrm{T}v, \mathrm{T}w \rangle = \langle v, w \rangle, \ \forall v, w \in V$$

- Orthogonal matrix:Orthogonal matrix: A square matrix $A$A is called an orthogonal matrix if $AA^T = A^T A = I$AAT = AT A = I(identity matrix)

*1 point*

Consider the following linear transformations $T_1, T_2 : R^2 \rightarrow R^2$, and $T_3, T_4 : R^3 \rightarrow R^2$T1, T2 : R2 → R2, and T3, T4 : R3 → R2 defined as

$$T_1(x, y) = (\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}), \text{T1(x, y) = (} \quad 2$$

√

x+y, 2

√

x−y),

$$T_2(x, y) = (x + 3y, 2x - 2y), \text{T2(x, y) = (x + 3y, 2x − 2y),}$$

$$T_3(x, y, z) = (\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}), \text{T3(x, y, z) = (} \quad 2$$

√

x+z, 3

√

x+y 3

√

+z),

$$T_4(x, y, z) = (x + y + z, x + y), \text{T4(x, y, z) = (x + y + z, x + y),}$$

where $R^3$R3 and $R^2$R2 are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

○

$T_1$T1

○

$T_2$T2

○

$T_3$T3

○

$T_4$T4

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T_1$T1

*1 point*

Which of the following option(s) is (are) true?

☐

[1 0 0−1][100−1] is an orthogonal matrix.

☐

$\frac{1}{\sqrt{2}}$[1−1 1 1] 2

√

1 [11−11] is an orthogonal matrix.

☐

$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & -2 & 1 \end{bmatrix}$ is an orthogonal matrix

☐

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ 2

$\sqrt{1}$ $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$ 2

$\sqrt{}$

is an orthogonal matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & -1 \end{bmatrix}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ 2

$\sqrt{}$

1 $\begin{bmatrix} 1 & 1 & -1 & 1 \end{bmatrix}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ 2

$\sqrt{}$

1 $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$ 2

$\sqrt{}$

is an orthogonal matrix.